

Final Exam Solutions:

1. (Counting; 15 pts)

In this problem, you may assume that "at random" or "in a random order" implies equally likely outcomes.

- (a) If 4 marbles are thrown at random into 10 boxes, what is the probability that no box receives more than one marble?
 - (b) Suppose that the letters $\{s, s, s, t, t, t, i, i, a, c\}$ are arranged in a random order. Compute the probability that they spell the word "statistics".
 - (c) Five cards are dealt at random from a deck of 52 cards. As usual, the deck is comprised of four distinct suits, each of which has the same 13 denominations (2, 3, ..., 10, J, Q, K, A). Compute the probability of obtaining a hand with all five cards from the same suit.
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1a) Counting:

of ways 4 marbles can be placed into 10 bins:

$$= 10 \times 10 \times 10 \times 10 = 10^4$$

↑ # bins possible for marble 1

of ways to place 4 marbles with no more than 1 per box:

$$= 10 \times 9 \times 8 \times 7 = 5040$$

↑ # possibilities for 1st marble
↑ # poss. for 2nd marble - can't be placed in same place as the 1st

$$\text{So: } IP(\text{no box has more than 1 marble}) = \frac{5040}{10^4} = \boxed{50.4\%}$$

1b) Counting:

Sub-exp. 1: Place the 3 "s" in any of 10 spots $C(10, 3)$

Sub-exp. 2: Place the 3 "t" in any of $10-3=7$ remaining spots $C(7, 3)$

Sub-exp. 3: Place the 2 "i" in any of $10-3-3=4$ spots $C(4, 2)$

Sub-exp. 4: Place the 1 "a" in any of $10-3-3-2=2$ spots $C(2, 1)$

Sub-exp. 5: Place the 1 "c" in any of $10-3-3-2-1=1$ spots $C(1, 1) = 1$

Total # Outcomes:

$$= C(10,3) \cdot C(7,3) \cdot C(4,2) \cdot C(2,1)$$

→ Only in 1 outcome is "statistics" spelled

$$\text{So: } P(\text{"statistics" is spelled}) = \frac{1}{C(10,3) \cdot C(7,3) \cdot C(4,2) \cdot C(2,1)} = \boxed{\frac{1}{50,400}}$$

1c) Counting:

Sub-exp. 1: Choose 1 out of 4 possible Suits $C(4,1)$

Sub-exp. 2: Choose 5 out of 13 denominations (for a given suit) $C(13,5)$

hands with 5 cards from the same suit:

$$= 4 \cdot C(13,5)$$

total possible 5-card hands:

$$= C(52,5)$$

So:

$$P(5 \text{ cards of same suit}) = \frac{4 \cdot C(13,5)}{C(52,5)} = \boxed{\frac{33}{16660}}$$

2. (COVID Antigen Tests; 15 pts)

During the COVID-19 pandemic several years ago, several tests were developed to determine whether individuals had the virus. Let “Test+” and “Test−” denote the events that an individual tested positive or negative, respectively, and let “COVID+” and “COVID−” denote the events that an individual had COVID or not, respectively.

It is estimated that 2.4% of the population had COVID during the peak in early 2022. One such antigen test has the following characteristics reported by the CDC:

- $\mathbb{P}(\text{Test} + | \text{COVID}+) = 65.3\%$
- $\mathbb{P}(\text{Test} - | \text{COVID}-) = 99\%$

In this problem you will analyze how the characteristics of this test affect an individual's likelihood of having (or not having) the disease conditioned on a positive (or negative) test result.

- Compute $\mathbb{P}(\text{Test}+)$ and $\mathbb{P}(\text{Test}-)$.
- Compute $\mathbb{P}(\text{COVID} + | \text{Test}+)$ and $\mathbb{P}(\text{COVID} - | \text{Test}-)$ for an individual chosen uniformly at random from the population.
- We would like to improve this test by increasing $\mathbb{P}(\text{Test} + | \text{COVID}+)$ to some new value $p \in (0, 1)$. What is the smallest value of p for which $\mathbb{P}(\text{COVID} + | \text{Test}+) \geq 70\%$?
Note: it is still the case that 2.4% of the population is COVID-positive, and the probability of COVID-negative individuals testing negative remains 99%. You will need to compute how $\mathbb{P}(\text{Test}+)$ changes with p .
- For the value of p you selected in part (c), what is $\mathbb{P}(\text{COVID} - | \text{Test}-)$?

$$\begin{aligned} 2a) \quad \mathbb{P}(\text{Test} +) &= \mathbb{P}(\text{Test} + | \text{COVID}+) \cdot \mathbb{P}(\text{COVID}+) \\ &\quad + \mathbb{P}(\text{Test} + | \text{COVID}-) \cdot \mathbb{P}(\text{COVID}-) \end{aligned} \quad \begin{array}{l} \text{(by Law of Total} \\ \text{Probability)} \end{array}$$

$$\begin{aligned} &= 65.3\% \cdot \underbrace{2.4\%}_{\substack{\uparrow \\ \mathbb{P}(\text{COVID}+)}} + (1 - 99\%) \cdot \underbrace{(1 - 2.4\%)}_{\substack{\uparrow \\ \mathbb{P}(\text{COVID}-) = 1 - \mathbb{P}(\text{COVID}+)}} \end{aligned}$$

$$\boxed{\mathbb{P}(\text{Test} +) = 2.54\%}$$

$$\mathbb{P}(\text{Test} -) = 1 - \mathbb{P}(\text{Test} +)$$

$$\boxed{\mathbb{P}(\text{Test} -) = 97.46\%}$$

2b) By Bayes' Rule:

$$P(C+|T+) = \frac{P(T+|C+) \cdot P(C+)}{P(T+)} = \frac{65.3\% \cdot 2.4\%}{2.54\%}$$

$$\boxed{P(C+|T+) = 61.6\%}$$

Likewise:

$$P(C-|T-) = \frac{P(T-|C-) \cdot P(C-)}{P(T-)} = \frac{99\% (1-2.4\%)}{97.46\%}$$

$$\boxed{P(C-|T-) = 99.15\%}$$

2c) Let $p = P(T+|C+)$

$$\text{Then: } P(T+) = p \cdot 2.4\% + (1-99\%)(1-2.4\%)$$

$$\text{And: } P(C+|T+) = \frac{p \cdot 2.4\%}{p \cdot 2.4\% + (1-99\%)(1-2.4\%)}$$

For this to be $\geq 70\%$,
need $\boxed{p \geq 94.9\%}$

2d) By Bayes' Rule:

$$P(C-|T-) = \frac{P(T-|C-) P(C-)}{P(T-)}$$

$$\downarrow P(T-) = P(T-|C-) P(C-) + P(T-|C+) P(C+)$$

$$= \frac{99\% \cdot (1-2.4\%)}{99\% \cdot (1-2.4\%) + (1-94.9\%) \cdot 2.4\%}$$

$$\boxed{P(C-|T-) = 99.99\%}$$

3. (MLE for Variance of a Normally Distributed Population; 15 pts)

Let $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$ be independent and identically distributed random variables from a Normal population with mean μ and variance σ^2 .

In this problem, we will construct the maximum likelihood estimator for the population variance σ^2 , assuming the population mean μ is known.

- Write the likelihood function $\mathcal{L}(\theta; X_1, \dots, X_n)$, and the log-likelihood function, where θ represents the population variance σ^2 . Recall that the population mean is μ , and is assumed to be known.
- Compute the maximum likelihood estimator $\hat{\theta}_{MLE}(X_1, \dots, X_n)$ for the population variance σ^2 .
- Is $\hat{\theta}_{MLE}(X_1, \dots, X_n)$ a biased estimator of σ^2 ? Justify your answer (a proof is not necessary, but is certainly acceptable).

$$3a) \mathcal{L}(\theta; X_1, \dots, X_n) = \prod_{i=1}^n f(X_i; \theta)$$

$f(\cdot; \theta)$ is the density of $N(\mu, \theta)$

$$\mathcal{L}(\theta; X_1, \dots, X_n) = (2\pi\theta)^{-\frac{n}{2}} \cdot \prod_{i=1}^n e^{-\frac{1}{2\theta}(X_i - \mu)^2}$$

$$\log(\mathcal{L}(\theta; X_1, \dots, X_n)) = -\frac{n}{2} \log(2\pi\theta) - \frac{1}{2\theta} \sum_{i=1}^n (X_i - \mu)^2$$

$$3b) \hat{\theta}_{MLE} = \underset{\theta}{\operatorname{argmax}} \log(\mathcal{L}(\theta; X_1, \dots, X_n))$$

$$0 = \frac{d}{d\theta} \log(\mathcal{L}(\hat{\theta}_{MLE}; X_1, \dots, X_n))$$

$$0 = -\frac{n}{2} \cdot \frac{1}{\hat{\theta}_{MLE}} + \frac{1}{2(\hat{\theta}_{MLE})^2} \cdot \sum_{i=1}^n (X_i - \mu)^2$$

$$n \cdot \hat{\theta}_{MLE} = \sum_{i=1}^n (X_i - \mu)^2$$

$$\hat{\theta}_{MLE} = \frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2$$

$$3c) \boxed{No} \text{ Since } \mu = E[X_i], \text{ then } E[\hat{\theta}_{MLE}] = E\left[\frac{1}{n} \sum_{i=1}^n (X_i - E[X_i])^2\right]$$

$$= \frac{1}{n} \cdot \sum_{i=1}^n E[(X_i - E[X_i])^2]$$

$= \operatorname{Var}(X_i)$, by def.

$= \sigma^2$ since $X_i \sim N(\mu, \sigma^2)$

4. (Confidence Interval for Exponential Parameter; 15 pts)

Let $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Exp}(\lambda)$, for some unknown parameter $\lambda > 0$. Assume that $n > 40$.

In this problem, we will build several distinct $1 - \alpha$ confidence intervals for the parameter λ .

- (a) Let $\bar{X}_n$ denote the mean, and $s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2$ denote the estimated variance. For any given $\alpha \in (0, 1)$, find a constant c for which:

$$\mathbb{P} \left(-c \leq \frac{\sqrt{n} (\bar{X}_n - \frac{1}{\lambda})}{s} \leq c \right) = 1 - \alpha.$$

- (b) Use the expression in part (a) to construct a two-sided $1 - \alpha$ confidence interval for λ . Be careful to notice that λ is **not** the mean of $X_1, \dots, X_n$.
- (c) Build a lower one-sided $1 - \alpha$ confidence interval for λ . Be sure to begin with a probabilistic statement (as in part (a)), justify it, and show your work in computing the confidence interval.

4a) Note: $E[X_i] = 1/\lambda$

So by CLT, $\frac{\sqrt{n}(\bar{X}_n - \frac{1}{\lambda})}{s(X_1, \dots, X_n)} \sim N(0, 1)$ since $n > 40$.

Hence, $\boxed{c = z_{\alpha/2}}$, by i) definition of critical values and ii) symmetry of std. norm. pdf.

$$4b) \mathbb{P} \left(-z_{\alpha/2} \leq \frac{\sqrt{n}(\bar{X}_n - \frac{1}{\lambda})}{s(\dots)} \leq z_{\alpha/2} \right) = 1 - \alpha$$

↓ Re-arranging

$$\mathbb{P} \left(\bar{X}_n - \frac{z}{\sqrt{n}} z_{\alpha/2} \leq \frac{1}{\lambda} \leq \bar{X}_n + \frac{z}{\sqrt{n}} z_{\alpha/2} \right) = 1 - \alpha$$

↓ Take to -1 power

$$\mathbb{P} \left(\lambda \in \left[\frac{1}{\bar{X}_n + \frac{z}{\sqrt{n}} z_{\alpha/2}}, \frac{1}{\bar{X}_n - \frac{z}{\sqrt{n}} z_{\alpha/2}} \right] \right) = 1 - \alpha$$

So: $\boxed{\left[\frac{1}{\bar{X}_n + \frac{z}{\sqrt{n}} z_{\alpha/2}}, \frac{1}{\bar{X}_n - \frac{z}{\sqrt{n}} z_{\alpha/2}} \right]}$ is a $1 - \alpha$ CI for λ

$$4c) \mathbb{P} \left(-z_{\alpha} \leq \frac{\sqrt{n}(\bar{X}_n - \frac{1}{\lambda})}{s} \right) = 1 - \alpha, \text{ by CLT \& definition of critical values}$$

↓ Re-arranging

$$\mathbb{P} \left(\frac{1}{\lambda} \leq \bar{X}_n + \frac{z}{\sqrt{n}} z_{\alpha} \right) = 1 - \alpha$$

So: $\boxed{\left[\frac{1}{\bar{X}_n + \frac{z}{\sqrt{n}} z_{\alpha}}, \infty \right)}$ is a lower $1 - \alpha$ CI for λ

5. (Cleanup Catastrophe; 15 pts)

Two siblings, Alice and Bob, have recently decided to draw lots to determine who is responsible for cleaning up after dinner. Alice, being the oldest, claimed to have a 'fair' jar that assigns $1/2$ probability to each of them. However, Bob has begun to question whether Alice is telling the truth...

Let $X_i = \begin{cases} 1 & \text{if Alice was selected to clean up on day } i \\ 0 & \text{otherwise.} \end{cases}$, and let $p = \mathbb{P}(X_i = 1)$.

In this problem, you will test the hypotheses $H_0 : p = 1/2$ and $H_A : p < 1/2$. You may assume that all X_i 's are independent and identically distributed.

- (a) Let $Y = \sum_{i=1}^{10} X_i$ denote the number of times Alice has been assigned the dishes in the first 10 days. What is the distribution of Y under $H_0 : p = 1/2$?
- (b) In these 10 days, Alice has only been assigned to do the dishes twice, i.e. we observe the realization $y = 2$. What is the probability of having seen $Y \leq 2$ under H_0 ?

Determine if we can reject $H_0 : p = 1/2$ at 5% significance level from this information.

Hint: is the probability calculated in this part a p-Value?

- (c) Bob would like to re-assess whether this procedure is fair after 60 days. Construct a hypothesis test of size 5% using $X_1, \dots, X_{60}$ for the specified null and alternative hypotheses. Be sure to: i) specify a test statistic, ii) construct an appropriate rejection region.

Note that the sample size here is $n = 60 > 40$.

- (d) After these 60 days, Alice has been chosen 27 times. I.e., we observed $\frac{1}{60} \sum_{i=1}^{60} x_i = 0.45$, and hence $s^2(x_1, \dots, x_n) = 0.252$. Determine whether or not these data $x_1, \dots, x_{60}$ are sufficient to reject H_0 at 5% significance level, and compute the p-Value.

5a) Since $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Ber}(\frac{1}{2})$ under H_0 , then their sum $Y = \sum_{i=1}^{10} X_i$ satisfies

$$Y \sim \text{Binomial}(10, \frac{1}{2})$$

5b) $\mathbb{P}(Y \leq 2 | H_0) = F_Y(2)$, where F_Y is the CDF of a $\text{Bin}(10, 1/2)$ R.V.

$$\mathbb{P}(Y \leq 2 | H_0) = 5.47\% \quad \leftarrow \text{This is a p-Value!}$$

Since the probability of seeing Y be as or more extreme ^(for $H_A: p < 1/2$) than Z under H_0 is larger than 5%, then we do not reject H_0 at 5% significance level.

$$5c) T(X_1, \dots, X_{60}) = \frac{\sqrt{60}(\bar{X}_{60} - 1/2)}{S(X_1, \dots, X_{60})} \leftarrow \text{Has } N(0,1) \text{ dist'n under } H_0$$

$$R = (-\infty, -z_{5\%}] \leftarrow P(T \in R | H_0) = 5\%, \text{ by def'n of critical values}$$

$$\text{So: } \mathcal{H}(X_1, \dots, X_n) = \begin{cases} 1 & \text{if } \frac{\sqrt{60}(\bar{X}_{60} - 1/2)}{S(X_1, \dots, X_{60})} \leq -1.63 \\ 0 & \text{o/w} \end{cases} \quad \text{is a test of size } 5\%$$

5d) Compute the value of our test statistic:

$$T(x_1, \dots, x_{60}) = \frac{\sqrt{60}(0.45 - 1/2)}{\sqrt{0.2521}} = -0.775 \notin R$$

$\leadsto$ Do not reject H_0

$$p\text{-Value} = \Phi(-0.775) \approx 22\%$$

6. (Comparing Power of Tests; 15 pts)

Let $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Unif}(0, \theta)$, where the parameter θ is unknown. Let the hypotheses of interest be $H_0 : \theta = \theta_0$, and $H_A : \theta > \theta_0$. Assume $n > 30$.

In this problem, you will construct and evaluate two hypotheses tests of size α based on different test statistics.

(a) Consider the test statistic:

$$T_1(X_1, \dots, X_n) = \frac{\sqrt{n}(\bar{X}_n - \frac{1}{2}\theta_0)}{\sigma},$$

where $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$ and $\sigma^2 = \text{Var}(X_i)$ is (unrealistically...) assumed to be known.

Construct a hypothesis test of size α for the hypotheses above using the test statistic $T_1(\dots)$. To obtain full credit, you must: i) specify a rejection region, and ii) show that under H_0 , the test statistic $T_1(\dots)$ falls in the rejection region with probability α .

(b) Compute the probability of a Type II Error for the test in part (a) when $\theta > \theta_0$, and H_A is true.

Hint: this should depend on θ , and should include the standard normal CDF.

(c) Consider another test statistic:

$$T_2(X_1, \dots, X_n) = \frac{X_{(n)}}{\theta_0},$$

where $X_{(n)} = \max_i X_i$. Find the CDF of $T_2(\dots)$ under H_0 , and identify a constant c for which:

$$\mathbb{P}(T_2(X_1, \dots, X_n) \geq c \mid H_0) = \alpha,$$

hence obtaining a rejection region of $[c, \infty)$ for another hypothesis test of size α , now based on the test statistic $T_2(\dots)$.

(d) Compute the probability of a Type II Error for the test in part (c) when $\theta > \theta_0$, and H_A is true.

Hint: this should also depend on θ , but should not include the standard normal CDF.

(e) Let $n = 36$ and $\sigma = 30$. For $\theta_0 = 100$, $\theta = 102$, and $\alpha = 5\%$, compute the probability of a Type II Error for each test, and identify the test with greatest power (i.e. lowest probability of Type II Error).

6a) Under H_0 , since $E[X_i] = \frac{\theta_0}{2}$, by the CLT,

$$T_1 \sim N(0, 1), \text{ under } H_0$$

So, $\mathbb{P}(T_1(X_1, \dots, X_n) \geq z_\alpha \mid H_0) = \alpha$, by def'n of z_α

$$\leadsto \boxed{R = [z_\alpha, \infty)}$$

6b) When $\theta > \theta_0$, then $T_2(\dots)$ is not $N(0,1)$

But:

$$T_2(X_1, \dots, X_n) = \frac{\sqrt{n}(\bar{X}_n - \frac{1}{2}\theta_0)}{\sigma} = \underbrace{\frac{\sqrt{n}(\bar{X}_n - \frac{1}{2}\theta)}{\sigma}}_{N(0,1) \text{ since } E[X_i] = \frac{\theta}{2}} + \frac{\frac{1}{2}\sqrt{n}(\theta - \theta_0)}{\sigma}$$

$$\Rightarrow T_2(\dots) \sim N\left(\frac{\frac{1}{2}\sqrt{n}(\theta - \theta_0)}{\sigma}, 1\right) \text{ when true parameter is } \theta$$

$$P(\text{Type II Error}) = P(T_2(\dots) \notin R \mid \theta > \theta_0)$$

$$= P\left(\frac{\sqrt{n}(\bar{X}_n - \frac{1}{2}\theta)}{\sigma} + \frac{\frac{1}{2}\sqrt{n}(\theta - \theta_0)}{\sigma} < z_\alpha \mid \theta > \theta_0\right)$$

$$= P\left(\underbrace{\frac{\sqrt{n}(\bar{X}_n - \frac{1}{2}\theta)}{\sigma}}_{\sim N(0,1)} < z_\alpha - \frac{\frac{1}{2}\sqrt{n}(\theta - \theta_0)}{\sigma} \mid \theta > \theta_0\right)$$

$$\boxed{P(\text{Type II Error}) = \Phi\left(z_\alpha - \frac{\frac{1}{2}\sqrt{n}(\theta - \theta_0)}{\sigma}\right)}$$

6c) CDF of $X_{(n)}$ under H_0 is: $F_{X_{(n)}}(x) = P(X_{(n)} \leq x) = \left(\frac{x}{\theta_0}\right)^n, x \in [0, \theta_0]$

$$S_0: P(T_2 \leq t) = P(X_{(n)} \leq t \cdot \theta_0) = F_{X_{(n)}}(t \cdot \theta_0)$$

$$\boxed{P(T_2 \leq t \mid H_0) = t^n}$$

$$P(T_2 \geq c \mid H_0) = \alpha$$

$$\Leftrightarrow P(T_2 \leq c \mid H_0) = 1 - \alpha$$

$$c^n = 1 - \alpha$$

$$\therefore \boxed{c = (1 - \alpha)^{1/n}}$$

$$6d) P(\text{Type II Error}) = P(T_2 < (1 - \alpha)^{1/n} \mid \theta > \theta_0)$$

$$= P\left(\frac{X_{(n)}}{\theta_0} < (1 - \alpha)^{1/n} \mid \theta > \theta_0\right)$$

$$= P\left(\underbrace{X_{(n)}}_{\text{Has CDF of } F_{X_{(n)}}(x) = \left(\frac{x}{\theta}\right)^n} < \theta_0 (1 - \alpha)^{1/n} \mid \theta > \theta_0\right)$$

Has CDF of $F_{X_{(n)}}(x) = \left(\frac{x}{\theta}\right)^n$

$$= \left(\frac{\theta_0 (1 - \alpha)^{1/n}}{\theta}\right)^n$$

$$\boxed{P(\text{Type II Error}) = \left(\frac{\theta_0}{\theta}\right)^n (1 - \alpha)}$$

6e) T_1 :

$$z_{5\%} = 1.65$$

$$P(\text{Type II Error}) = \Phi\left(1.65 - \frac{\sqrt{36}(100-102)}{30}\right)$$

$$= \Phi(1.65 - 0.2)$$

$$= 92.5\%$$

T_2 :

$$P(\text{Type II Error}) = \left(\frac{100}{102}\right)^{36} (1 - 5\%)$$

$$= 46.6\%$$

The second test has higher power